# Supplement A - Technical Details for The Link Function Problem in Psychological Interaction Testing

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## S1. Scope of this supplement

This supplement reports extended review methods and reliability, followed by the chance-corrected-link algebra and the generative parameters behind the illustrative figures. It then gives full specifications and estimation details for the three worked simulations and the diagnostic simulation, together with supplementary results.

## S2. Preregistered review: extended methods and reliability

### Protocol and sampling frame

The review protocol was preregistered before any record was screened and is available at https://osf.io/bdu73/, together with the coding data dictionary. The protocol specified a descriptive review of recent psychological research articles from five pre-specified journals, one per major area: *Psychological Science* (general), *Journal of Experimental Psychology: General* (experimental), *Journal of Personality and Social Psychology* (social), *Developmental Science* (developmental), and *Psychological Medicine* (clinical). Journals were selected on a discretionary combination of criteria stated in the protocol: first-quartile Impact Factor in the 2023 Journal Citation Reports, not primarily publishing reviews, not narrowly specialized, and strong general reputation in the respective field, as jointly evaluated by the coauthors.

Article records were retrieved from Elsevier's Scopus database for all articles published in these journals in 2024. For each journal, the full record list was randomly ordered and then screened in that order until 40 eligible interaction-testing articles were included, for a planned total of 200 interaction-testing articles across the five journals. Screening a larger set also allowed estimating the proportion of eligible empirical articles that test interactions. The design is targeted and descriptive: it characterizes reporting practice in influential outlets during the sampled year. It is not a random or stratified sample of all psychology journals, and the estimates are not prevalence estimates for the discipline.

### Eligibility and coding workflow

Eligible articles were empirical articles presenting original psychological research. The following were excluded, per protocol: reviews and meta-analyses; editorials, commentaries, opinion papers, and theoretical articles without empirical data; qualitative-only articles; replication-only studies; and psychometric validation studies. Articles were coded in detail only if they tested at least one interaction, through statistical modeling or ANOVA. The coding scheme targeted observed-outcome analyses; articles whose relevant tests were conducted only at the latent-variable level, or whose empirical analyses relied only on structural equation models, were excluded from detailed coding.

The resulting screening flow is reported in Table 1.

**Table 1**
*Screening flow for the preregistered review.*

| Step | N |
|---|---|
| records retrieved across journal sheets | 2175 |
| eligible empirical articles in final dataset | 331 |
| eligible articles testing at least one interaction | 200 |
| eligible articles not testing interactions | 131 |
| note-based SEM / latent-variable-only exclusions | 12 |

Each interaction-testing article was independently coded by two researchers. Discrepancies were resolved through discussion or, if needed, by consulting the other coauthors. Generative AI tools were used only as an optional aid to locate relevant passages within an article; all coding used in the review is human.

**Table 2**
*Coded variables and their preregistered definitions.*

| Variable | Definition (preregistered) | Type |
|---|---|---|
| Tests interactions | Whether the article tests at least one interaction term in a statistical model or using an ANOVA. | Boolean |
| Uses non identity link function | Whether at least one tested interaction is presumably analyzed with a non-identity link function (for example, because a generalized linear model was used). | Boolean |
| Explicit link function | Whether the link function is explicitly declared for at least one statistical model in the article. Indicating only which generalized linear model or family was used is not enough. | Boolean |
| Incorrect identity link function | Whether at least one tested interaction is analyzed using an identity link function when another, non-identity link function would be more appropriate. | Boolean |
| Finds significant interactions | Whether at least one interaction, among those where the link was coded as incorrectly identity, yields a statistically significant result, or is considered and discussed analogously in non-frequentist analyses. | Boolean |
| Response variable types | Non-exclusive list of response-variable types on which interactions are tested (for example, sum score, accuracy, error count, response times, ordinal/Likert). | Category |

Beyond these definitions, the coders applied a small set of operational rules for the recurring borderline decisions. Table 3 records the ones that most affect the reported counts.

**Table 3**

*Operational rules the coders applied for the decisions that most affect the reported counts, condensed from the coding sheet. The preregistered variable definitions are in Table 2.*

| Variable | Coded "Yes" when | Decisions that most affect the count |
|---|---|---|
| Non-identity link used | at least one tested interaction is fitted with a model implying a non-identity link, that is, any GLM or GLMM beyond the Gaussian-identity default (ANOVA, ordinary regression, linear mixed model). | A model named "logistic", "probit", "Poisson", "negative binomial", "ordinal"/"cumulative", or "Gamma" regression implies a non-identity link. |
| Explicit link function | the link is identifiable from the report, either stated in words or named through a link-specific model. | "Logistic regression" (logit), "probit regression" (probit), and "Poisson" or "log-linear" (log) count as explicit. Naming only a family, a "generalized linear model", or a "GLMM" does not. |
| Gaussian-identity analysis on a constrained outcome | an interaction is tested with an identity-link model (ANOVA, ordinary regression, linear mixed model, or a test on cell means) applied to an outcome with formal constraints: bounded, discrete, positive-only, or with a known chance level. | Flagged only when the constraint is clear from the report. Not flagged when the analysis is explicitly justified as an observed-score estimand, or when the outcome is plausibly continuous and far from any bound. |
| Significant interaction | at least one interaction already flagged as a Gaussian-identity analysis on a constrained outcome is reported as statistically significant, or is interpreted analogously (for example, a credible interval excluding zero) in a Bayesian analysis. | Restricted to interactions already flagged in the row above. |
| Response-variable types | — | Non-exclusive: one article can contribute several types (sum score, accuracy or proportion, error count, response time, ordinal/Likert, neural or physiological, difference score, correlation). |

One additional decision beyond those summarized in Table 3 concerned transformed outcomes. Coding referred to the response on the scale on which it entered the model. When authors transformed the outcome before applying a linear model or ANOVA, the analysis was evaluated on the transformed scale and was not flagged. For example, a linear model or ANOVA applied directly to raw response times was flagged because response times are constrained to positive values, whereas the same analysis applied to log transformed response times was not. Logarithmic transformations were the most common, but the same rule was applied to the other transformations encountered in the reviewed articles.

Two notes clarify the interpretation of the coding variables. First, *Incorrect identity link function* is a conservative screening flag for a formal mismatch between the observed-outcome constraints and the identity-link generative model. Cases were flagged only when the constraint was identifiable from the published report; when the report did not allow this judgment, the case was not counted. The flag is not an error rate, not a severity grade, and does not imply that the substantive conclusion is false, since an observed-score estimand can be a deliberate and defensible target. Second, *Response variable types* is non-exclusive: one article can contribute to several categories, so the category counts do not sum to the number of articles.

The last coding variable is described in greater detail in Table 4, which summarizes the response variable labels used during coding and their relation to the broader categories reported in the main paper. Classification was based primarily on the process that generated the response rather than on the terminology used by the article authors. When this process could not be determined reliably from the report, the authors' terminology was retained.

**Table 4**

*Operational definitions of the broad response variable categories reported in the main paper and their principal detailed coding labels. An article could contribute to more than one row.*

| Broad category in the main paper | Principal detailed labels | Operational definition and coding decisions |
| --- | --- | --- |
| Binary / accuracy / proportions | Binary; proportion | *Binary* indicated a finite number of discrete binary events, such as correct and incorrect responses over a predetermined number of trials. This label was retained when binary responses were summed or averaged. *Proportion* was reserved for responses treated as varying continuously between 0 and 1 and not determined by a fixed number of discrete binary events. Accuracy calculated over a predetermined number of trials was therefore coded as binary rather than as a continuous proportion. |
| Sum scores / composites | Sum score; log transformed sum score; normalized sum score | A *sum score* was a sum or average of multiple ordinal or slider responses. Aggregates of dichotomous responses were instead coded as binary. Log transformed and normalized totals were identified separately during detailed coding but were included in the same broad category for reporting. |

| Broad category in the main paper | Principal detailed labels | Operational definition and coding decisions |
|---|---|---|
| Ordinal / Likert / ratings | Ordinal; slider | *Ordinal* indicated a single ordered response with no more than 10 response levels. A single ordered response with more than 10 available levels was coded as a *slider*. When multiple ordinal or slider responses were summed or averaged, the resulting outcome was coded as a sum score. |
| Response times / durations | Response time; log transformed response time; inverse response time | Positive response or reaction times were included in this category whether they were analyzed on their original scale, after a logarithmic transformation, or after a reciprocal transformation. The specific transformation was retained in the detailed coding. |
| Neural / physiological measures | Amplitude; log transformed amplitude; heart rate; heart rate variability; skin conductance response | Continuous neural and physiological measurements were included in this category. When a physiological response represented a count rather than an amplitude or continuous measurement, it was classified according to the process that generated the count. |
| Difference / distance scores | Difference score; distance | A *difference score* was the arithmetic difference between two individual responses. *Distance* referred to continuous distance measures, including angular deviation, positional error, and geodesic distance. |

| Broad category in the main paper | Principal detailed labels | Operational definition and coding decisions |
|---|---|---|
| Counts / error counts | Count; log transformed count; frequency | *Count* was reserved for counts without a predetermined upper bound. Counts of correct responses or errors over a fixed number of trials were coded as binary because the number of possible events was predetermined. Generic descriptions such as “errors” were therefore classified according to the underlying process whenever it could be established from the report. |
| Correlations / associations | Correlation; Fisher z transformed correlation | Correlations used as outcomes and their Fisher z transformations were distinguished during detailed coding and combined in this broad category for reporting. |

Coding relied only on what a reader can observe in a published report: the analysis described, the outcome described, and any explicit statement of family or link. The review therefore documents reporting practice, not the correctness of substantive conclusions. For the same reason, the review did not classify interactions as removable or nonremovable at the article level, because published reports rarely provide the target contrast, the complete coefficients, and the assumed measurement scale that such a classification requires.

**Interrater agreement**

Double-coded records with complete coder pairs are summarized in Table 5, with raw percentage agreement and Cohen’s kappa reported together. Several coded variables had imbalanced marginal distributions, a setting in which kappa can be conservative, so the coder-specific yes rates are reported alongside to let readers evaluate the agreement pattern directly.

**Table 5**
*Interrater agreement for double-coded records.*

| Item | N | Agreement | Kappa | Coder 1 yes | Coder 2 yes |
|---|---|---|---|---|---|
| Uses non-identity link | 199 | 92.0% | 0.81 | 28.1% | 30.2% |
| Explicit link reported | 199 | 88.9% | 0.66 | 20.6% | 20.6% |
| Gaussian-identity flagged | 197 | 88.3% | 0.71 | 71.6% | 72.1% |
| Significant flagged interaction | 141 | 91.5% | 0.68 | 86.5% | 82.3% |

## By-journal descriptives

Table 6 reports the review descriptives by journal. Percentages are computed within interaction-testing articles, except for the interaction-testing percentage, which is computed within eligible articles. The main paper reports only the range of these quantities across journals and the position of *Psychological Medicine*, which had the lowest screening proportion and the highest non-identity-link proportion in this sample.

**Table 6**
*By-journal review descriptives.*

| Journal | Eligible | Interaction | Interaction / eligible | Non-identity link | Explicit link | Gaussian-identity flagged |
|---|---|---|---|---|---|---|
| Developmental Science | 63 | 40 | 63.5% | 25.0% | 17.5% | 57.5% |
| JEP: General | 47 | 40 | 85.1% | 27.5% | 27.5% | 75.0% |
| JPSP | 55 | 40 | 72.7% | 25.0% | 15.0% | 92.5% |

| Journal | Eligible | Interaction | Interaction / eligible | Non-identity link | Explicit link | Gaussian-identity flagged |
|---|---|---|---|---|---|---|
| Psychological Medicine | 105 | 40 | 38.1% | 40.0% | 35.0% | 40.0% |
| Psychological Science | 61 | 40 | 65.6% | 27.5% | 20.0% | 95.0% |

## S3. The chance-corrected binomial link

For binomial accuracy with a known chance level $c$, the chance-corrected link maps the linear predictor onto the interval from $c$ to 1,

$$p_i = c + (1 - c)\, F(\eta_i),$$

where $F$ is a cumulative distribution function, taken to be logistic in the analyses reported here. The corresponding link applies $F^{-1}$ to the above-chance quantity,

$$g(p_i) = F^{-1}\left(\frac{p_i - c}{1 - c}\right).$$

For a two-alternative forced-choice or true/false task with $c = .50$ this becomes $p_i = .50 + .50\, F(\eta_i)$. A standard binomial logit or probit is the special case $c = 0$. Both specifications respect the binomial sampling process, but they define different no-interaction baselines: a product term under the standard link tests additivity over the full 0-to-1 range, whereas under the chance-corrected link it tests additivity on a performance-above-chance scale that already encodes the task-implied lower asymptote.

The fixed-chance form used in the paper is the simplest member of a broader psychometric-function family. A more general version adds a lapse parameter $\lambda$ that lowers the upper asymptote,

$$p_i = c + (1 - c - \lambda)\, F(\eta_i),$$

so that performance saturates below 1 when occasional lapses are plausible (Knoblauch & Maloney, 2012; Wichmann & Hill, 2001). The analyses in the paper fix $\lambda = 0$ and fix $c$ by task design, for transparency and identifiability; the lapse-rate version is the natural extension when the design and data can support estimation of an upper asymptote.

## S4. Data-generating processes and estimation details

This section gives the formal generative models and the estimation internals for the simulations. The main paper describes the purpose and interpretation of each simulation; here we state the equations, the fixed parameter values, and the fitting procedures needed to reproduce the reported behaviour. Unless noted otherwise, each scenario used a significance level of $\alpha = .05$, and each simulation set a fixed random seed.

### Generative parameters of the illustrative figures

Two figures in the main paper are illustrations rather than simulation results, so their parameters are recorded here.

The introductory count example samples $N = 250$ children with age drawn uniformly between 6 and 10 and group assigned by a fair coin. Error counts are Poisson with a log link,

$$\log \mu_i = 2.60 - 0.55\,(\text{age}_i - 6) + 0.65\,\text{group}_i,$$

with no age-by-group product term on the log scale. The figure plots a Poisson-log model fitted to one such sample, on the count scale and on the log scale. The count scenario of the diagnostic simulation (D1) adapts this coefficient pattern rather than reproducing it exactly: it keeps the same generating structure but uses a smaller sample ($N = 180$), a slightly lower intercept (2.20), and a smaller group effect (0.45); its full specification is given below under *Diagnostic simulation: data-generating processes*.

The motivating $2 \times 2$ figure is fully deterministic: it evaluates four cell probabilities, without sampling. They are generated additively on the chance-corrected logit scale with $c = .50$,

$$p = .50 + .50\,\text{logistic}(\eta), \qquad \eta = -1.00 + 2.10\,\text{condition} + 1.70\,\text{group},$$

again with no condition-by-group product term. The resulting cell probabilities are given in Table 7. Because a saturated $2 \times 2$ design has exactly one product-term coefficient per scale, those same four probabilities imply the three different interaction coefficients reported in Table 8: exactly zero on the generating scale, positive on the standard logit scale, and negative on the raw probability scale.

**Table 7**
*Cell probabilities of the motivating 2 x 2 figure. These four values are identical in all three panels of that figure.*

| Group | Condition | Linear predictor | Expected probability |
|---|---|---|---|
| Group 0 | 0 | -1.00 | 0.634 |
| Group 0 | 1 | 1.10 | 0.875 |
| Group 1 | 0 | 0.70 | 0.834 |
| Group 1 | 1 | 2.80 | 0.971 |

**Table 8**

*Product-term coefficient implied by the cell probabilities in the previous table, on each of the three scales shown in the motivating figure. Only the scale on which the difference between differences is computed changes.*

| Scale | Implied product term |
|---|---|
| Linear (probability) | -0.103 |
| Standard logit | 0.513 |
| Chance-corrected logit | 0.000 |

## Simulation 1: forced-choice accuracy with a chance floor

For each participant $i$, age is drawn on the range 6 to 10 and centered at 8, group is a two-level factor, and the number of correct responses out of $k = 20$ trials is binomial with

$$p_i = .50 + .50\,\text{logistic}(\eta_i), \qquad \eta_i = \beta_0 + 0.60\,\text{age}_c - 0.90\,\text{group} + 0\,(\text{age}_c \times \text{group}).$$

The generating scale is the chance-corrected logit with a fixed floor of .50, and the age-by-group product term is exactly zero on that scale. Three scenarios move the same structure along the accuracy range through the intercept: lower performance ($\beta_0 = -0.80$), middle performance ($\beta_0 = 0.00$), and higher performance ($\beta_0 = 0.80$). Each replication samples $N = 250$ participants.

Four candidate models are fitted to each replication with the same `age_c * group` interaction formula: (1) a Gaussian identity model on the accuracy proportion; (2) a standard binomial logit on successes out of trials; (3) a standard binomial probit; and (4) a chance-corrected binomial logit fitted by direct maximum likelihood on the binomial likelihood with mean $p = c + (1 - c)\,\text{logistic}(\eta)$ and $c = .50$ fixed by design. To avoid false convergence on the flat region of the likelihood near the chance floor, the chance-corrected fit uses three starting values: a zero vector, coefficients from the corresponding standard binomial-logit fit, and coefficients obtained by regressing the empirical above-chance proportions on the model matrix after logit transformation. Each start is optimized with BFGS using an analytic gradient, and the solution with the highest likelihood is retained. The observed Hessian is then computed at that solution and supplies the variance-covariance matrix for the Wald tests. A test is retained only when the optimizer converges, the Hessian is positive definite and sufficiently well conditioned, and the covariance matrix and Wald $p$-values are finite. Replications failing any of these checks are recorded as unusable fits. With this procedure, all four models returned usable tests in all 3,000 replications of each performance scenario.

**Table 9**

*Product-term rejection rates by forced-choice performance scenario and fitted model. The matched chance-corrected model shows nominal rejection; alternative scales show pseudo-interaction detection. Reject = rejected product-term tests; CI = Wilson 95%. All scenarios used 3,000 replications.*

| Scenario | Model | Rate type | Reject | Rejection rate [95% CI] |
|---|---|---|---:|---|
| Lower performance | Gaussian identity | Pseudo-interaction detection rate | 1306 | 43.5% [41.8, 45.3] |
| Lower performance | Standard binomial logit | Pseudo-interaction detection rate | 1787 | 59.6% [57.8, 61.3] |
| Lower performance | Standard binomial probit | Pseudo-interaction detection rate | 1716 | 57.2% [55.4, 59.0] |
| Lower performance | Chance-corrected binomial logit | Nominal rejection rate | 121 | 4.0% [3.4, 4.8] |
| Middle performance | Gaussian identity | Pseudo-interaction detection rate | 452 | 15.1% [13.8, 16.4] |
| Middle performance | Standard binomial logit | Pseudo-interaction detection rate | 1712 | 57.1% [55.3, 58.8] |
| Middle performance | Standard binomial probit | Pseudo-interaction detection rate | 1450 | 48.3% [46.5, 50.1] |
| Middle performance | Chance-corrected binomial logit | Nominal rejection rate | 131 | 4.4% [3.7, 5.2] |
| Higher performance | Gaussian identity | Pseudo-interaction detection rate | 416 | 13.9% [12.7, 15.1] |
| Higher performance | Standard binomial logit | Pseudo-interaction detection rate | 965 | 32.2% [30.5, 33.9] |

| Scenario | Model | Rate type | Reject | Rejection rate [95% CI] |
|---|---|---|---|---|
| Higher performance | Standard binomial probit | Pseudo-interaction detection rate | 572 | 19.1% [17.7, 20.5] |
| Higher performance | Chance-corrected binomial logit | Nominal rejection rate | 156 | 5.2% [4.5, 6.1] |

For the matched chance-corrected model, the Monte Carlo standard deviation of the estimated age-by-group coefficient was 0.180 in the lower-performance scenario, 0.115 in the middle-performance scenario, and 0.099 in the higher-performance scenario. Performance closer to the chance floor therefore provides less precise information about changes on the above-chance link scale.

**Simulation 2: within-family link choice**

Repeated binary trials are generated in a two-group, two-condition design with $n = 700$ participants, $k = 15$ trials per participant-condition cell, and a subject random intercept. The reference fixed effects on the probit scale are intercept 1.50, group $-1.00$, condition $-1.00$, and product term 0. When the generating link is logit, all fixed effects are multiplied by 1.65, so that the logit and probit data-generating processes imply similar cell probabilities while preserving a zero product term on the generating link scale. The random-intercept standard deviation is set from a latent-scale ICC of .30, using the latent residual variance of the generating link ($\pi^2/3$ for logit, 1 for probit).

Generating and fitted links are fully crossed (logit and probit in both roles), giving two matched-link and two wrong-link cells. All models are random-intercept GLMMs fitted with `glmmTMB`, using the same `group * condition` fixed-effect formula. The separate fitted-curve illustration uses a grouped-binomial dataset with 1,000 observational units and 60 trials per unit, with one `cbind(correct, incorrect)` response per unit; it is shown in Section S5, Figure 1.

**Table 10**

*Product-term rejection rates for the within-family link simulation. Matched links show nominal rejection; wrong links show pseudo-interaction detection. The median coefficient and SE are on the fitted link scale. CI = Wilson 95%. All scenarios used 3,000 replications.*

| Generating | Fitted | Status | Rate type | Reject | Rejection rate [95% CI] | Median beta | Median SE |
|---|---|---|---|---|---|---|---|
| logit | logit | Matched link | Nominal rejection rate | 137 | 4.6% [3.9, 5.4] | -0.000 | 0.075 |
| logit | probit | Wrong link | Pseudo-interaction detection rate | 504 | 16.8% [15.5, 18.2] | -0.041 | 0.043 |
| probit | logit | Wrong link | Pseudo-interaction detection rate | 543 | 18.1% [16.8, 19.5] | 0.077 | 0.076 |
| probit | probit | Matched link | Nominal rejection rate | 160 | 5.3% [4.6, 6.2] | -0.001 | 0.043 |

## Simulation 3: sum scores

Each replication samples $N = 600$ participants. A latent trait is generated as

$$\theta_i = 0.85\, x_i - 0.90\, \mathrm{group}_i + 0\,(x_i \times \mathrm{group}_i) + \varepsilon_i, \qquad \varepsilon_i \sim N(0, 1),$$

so the x-by-group product term is exactly zero on the latent generating scale. The latent trait is mapped to $J = 9$ ordinal items, each scored 0 to 3, through a graded logistic threshold model: item $j$ has three ordered thresholds, and the probability of reaching category $k$ or higher is logistic$(\theta_i - \tau_{jk})$. Thresholds are deterministic, with base values $(-1, 0, 1)$, evenly spaced item offsets from $-0.45$ to 0.45 across the nine items, and a scenario-level shift: middle of scale (shift 0.00), near floor (shift $+1.60$, harder items compress scores toward the lower bound), and near ceiling (shift $-1.60$). The observed outcome is the item total, bounded between 0 and 27, discrete, and differentially compressed across the score range.

Three analyses are fitted with the same `x * group` formula, to separate the relevant scales: (1) an observed sum-score identity model, `lm()` on the manifest total; (2) a Gaussian-probit bounded-score model, a Gaussian GLM with a probit link fitted to the continuity-corrected score proportion $(\text{sum} + 0.5)/(\text{max} + 1)$, included as a sensitivity model on the manifest total rather than a measurement model; and (3) a latent-scale benchmark, `lm()` on the true $\theta$, available only because $\theta$ is known in the simulation.

**Table 11**

*Product-term rejection rates by sum-score scenario and analysis scale. The latent generating-scale benchmark shows nominal rejection; observed-score models show pseudo-interaction detection. The latent benchmark is available only because theta is known. CI = Wilson 95%. All scenarios used 3,000 replications.*

| Scenario | Model | Rate type | Reject | Rejection rate [95% CI] |
|---|---|---|---|---|
| Lower range | Latent generating scale | Nominal rejection rate | 136 | 4.5% [3.8, 5.3] |
| Lower range | Observed sum-score identity | Pseudo-interaction detection rate | 2189 | 73.0% [71.3, 74.5] |
| Lower range | Gaussian-probit bounded score | Pseudo-interaction detection rate | 271 | 9.0% [8.1, 10.1] |
| Middle range | Latent generating scale | Nominal rejection rate | 138 | 4.6% [3.9, 5.4] |
| Middle range | Observed sum-score identity | Pseudo-interaction detection rate | 248 | 8.3% [7.3, 9.3] |
| Middle range | Gaussian-probit bounded score | Pseudo-interaction detection rate | 108 | 3.6% [3.0, 4.3] |
| Upper range | Latent generating scale | Nominal rejection rate | 133 | 4.4% [3.8, 5.2] |
| Upper range | Observed sum-score identity | Pseudo-interaction detection rate | 1108 | 36.9% [35.2, 38.7] |
| Upper range | Gaussian-probit bounded score | Pseudo-interaction detection rate | 140 | 4.7% [4.0, 5.5] |

**Diagnostic simulation: data-generating processes**

The main paper introduces the five diagnostic scenarios in prose. Their formal generating models are given here. In every scenario the product term is exactly zero on the generating link scale, and the fitted model keeps the outcome family and the interaction formula while changing only the link.

*D1, count errors (Poisson log, fitted with identity).* For $N = 180$ children with age uniform on $[6, 10]$ and a balanced group factor, errors are Poisson with

$$\log \mu_i = 2.20 - 0.55\,(\text{age}_i - 6) + 0.45\,\text{group}_i .$$

*D2, chance-floor accuracy (chance-corrected logit, fitted with standard logit).* The lower-performance scenario of Simulation 1: $N = 250$ participants, $k = 20$ trials per participant, and

$$p_i = .50 + .50\,\text{logistic}(\eta_i), \qquad \eta_i = -0.80 + 0.60\,\text{age}_{c,i} - 0.90\,\text{group}_i .$$

*D3, binary repeated trials (probit, fitted with logit).* A two-group, two-condition design with $N = 300$ participants, 15 trials per participant-condition cell, a subject random intercept, and a latent ICC of .30:

$$\Pr(y_{ij} = 1) = \Phi\big(1.50 - 1.00\,\text{group}_i - 1.00\,\text{condition}_j + u_i\big)\,.$$

*D4, binary with a continuous predictor (probit, fitted with logit).* The same structure as D3 — $N = 300$ participants, 30 trials each — with the two-level condition replaced by a continuous predictor $x \sim \text{Uniform}(0, 1)$, so that the contrast from $x = 0$ to $x = 1$ has the same probit-scale size as the condition contrast in D3. This scenario exists because residual checks over a focal predictor are better identified when the predictor has many unique values.

*D5, Gamma mean response time (log, fitted with inverse).* For $N = 220$ participants, response times are Gamma with shape 8 and

$$\log \mu_i = \log(400) + 0.25\, x_i + 0.25\, \text{group}_i,$$

so the baseline mean is 400 ms at $x = 0$ in the reference group.

Table 12 reports the expected values these parameters imply, which is what the misspecified link has to reproduce with a product term.

**Table 12**

*Expected values implied by the five diagnostic data-generating processes, at low, middle, and high values of the focal predictor. Values for the two binary scenarios are conditional on a subject random intercept of zero. Units differ by scenario and are given in the outcome column.*

| Scenario | Predictor | Value | Group | Linear predictor | Outcome | Expected |
|---|---|---|---|---|---|---|
| Count errors | age | 6.0 | Group 0 | 2.20 | expected count | 9.025 |
| Count errors | age | 8.0 | Group 0 | 1.10 | expected count | 3.004 |
| Count errors | age | 10.0 | Group 0 | 0.00 | expected count | 1.000 |
| Count errors | age | 6.0 | Group 1 | 2.65 | expected count | 14.154 |
| Count errors | age | 8.0 | Group 1 | 1.55 | expected count | 4.711 |
| Count errors | age | 10.0 | Group 1 | 0.45 | expected count | 1.568 |
| Chance-floor accuracy | age | 6.0 | Group 0 | -2.00 | expected accuracy | 0.560 |
| Chance-floor accuracy | age | 8.0 | Group 0 | -0.80 | expected accuracy | 0.655 |

| Scenario | Predictor | Value | Group | Linear predictor | Outcome | Expected |
|---|---|---|---|---|---|---|
| Chance-floor accuracy | age | 10.0 | Group 0 | 0.40 | expected accuracy | 0.799 |
| Chance-floor accuracy | age | 6.0 | Group 1 | -2.90 | expected accuracy | 0.526 |
| Chance-floor accuracy | age | 8.0 | Group 1 | -1.70 | expected accuracy | 0.577 |
| Chance-floor accuracy | age | 10.0 | Group 1 | -0.50 | expected accuracy | 0.689 |
| Binary repeated trials | condition | 0.0 | Group 0 | 1.50 | expected probability (u = 0) | 0.933 |
| Binary repeated trials | condition | 1.0 | Group 0 | 0.50 | expected probability (u = 0) | 0.691 |
| Binary repeated trials | condition | 0.0 | Group 1 | 0.50 | expected probability (u = 0) | 0.691 |
| Binary repeated trials | condition | 1.0 | Group 1 | -0.50 | expected probability (u = 0) | 0.309 |
| Binary continuous predictor | x | 0.0 | Group 0 | 1.50 | expected probability (u = 0) | 0.933 |
| Binary continuous predictor | x | 0.5 | Group 0 | 1.00 | expected probability (u = 0) | 0.841 |
| Binary continuous predictor | x | 1.0 | Group 0 | 0.50 | expected probability (u = 0) | 0.691 |
| Binary continuous predictor | x | 0.0 | Group 1 | 0.50 | expected probability (u = 0) | 0.691 |
| Binary continuous predictor | x | 0.5 | Group 1 | 0.00 | expected probability (u = 0) | 0.500 |
| Binary continuous predictor | x | 1.0 | Group 1 | -0.50 | expected probability (u = 0) | 0.309 |
| Gamma mean response time | x | -2.0 | Group 0 | 5.49 | expected mean (ms) | 242.612 |
| Gamma mean response time | x | 0.0 | Group 0 | 5.99 | expected mean (ms) | 400.000 |
| Gamma mean response time | x | 2.0 | Group 0 | 6.49 | expected mean (ms) | 659.489 |
| Gamma mean response time | x | -2.0 | Group 1 | 5.74 | expected mean (ms) | 311.520 |
| Gamma mean response time | x | 0.0 | Group 1 | 6.24 | expected mean (ms) | 513.610 |
| Gamma mean response time | x | 2.0 | Group 1 | 6.74 | expected mean (ms) | 846.800 |

**Diagnostic simulation: estimation and checks**

The diagnostic simulation ties each of the five link-misspecification scenarios defined above to a misspecified fitted link, holding the outcome family and the interaction formula fixed. In every replication the interaction model is fitted under the wrong link, and the primary diagnostics are applied to that same fitted model, because that is the model an analyst would inspect after finding the interaction. The same diagnostics are also applied to the corresponding correct-link interaction model as a calibration check. The technical specification of the checks is as follows.

*Simulation-based residuals* (Dunn & Smyth, 1996; Hartig, 2024) were computed with DHARMa, simulating 250 responses from the fitted model per replication. Four prespecified checks were applied: residual uniformity (`testUniformity`), dispersion (`testDispersion`), residual quantile patterns over fitted values (`testQuantiles`), and residual structure over the focal predictor. The last check is scenario-aware: `testQuantiles` over the predictor is used when the predictor has at least eight unique values; for the $2 \times 2$ repeated-trials scenario, residuals are instead compared across the group-by-condition design cells using a Kruskal-Wallis test on the DHARMa scaled residuals. Table 19 reports this design-cell check for D3. Cells are marked "–" only when a check is not applicable or returns no usable replications in the saved diagnostic summary.

*The Pregibon-style added-term link check* (Pregibon, 1980) extracts the fitted linear predictor, adds its square to the model formula, and tests the significance of the added term while retaining the family, link, base formula, and random-effects structure of the model being diagnosed. For GLMMs, the constructed term uses the population-level linear predictor, with the estimated random effects set to zero; this avoids feeding outcome-derived empirical Bayes estimates back into the model as an ordinary covariate. The check is reported as not applicable when the squared predictor does not increase the rank of the fixed-effects design matrix. In particular, the $2 \times 2$ interaction model is saturated over its four design cells, so its squared fitted linear predictor is exactly collinear with the existing fixed effects.

*The chance-corrected fits in D2* are the only ones estimated outside a standard model-fitting function, and they use an extended version of the same defensive direct-likelihood procedure used in Simulation 1, because here the target-link model is one of the two candidates in the AIC comparison and a failed fit would silently remove a replication from that comparison. The model is fitted by direct maximum likelihood with $c = .50$ fixed by design. Five starting values are tried: a vector of zeros, three derived from the corresponding standard-logit fit, and one from a linear fit on the empirical above-chance proportions, which is the scale the likelihood actually works on. Each start is optimized with both BFGS and Nelder–Mead, the solution with the highest likelihood is retained, and it is then polished with a tighter BFGS run. Because `optim()` reports only that a stopping rule was met, and that rule is easily met far from the maximum on the flat region near the chance floor, the retained solution is additionally checked against the score and the curvature of the log-likelihood; the same Hessian supplies the variance-covariance matrix used for the Wald test. Replications that fail these checks are recorded as failed fits.

*The same-formula AIC comparison* contrasts the target-link and the misspecified-link interaction models, holding the response data, the likelihood family, and the structural formula fixed, so only the mean-link mapping varies. No inconclusive ΔAIC category is used, so the two AIC rows in Table 19 are complementary whenever both AIC values are available.

## S5. Supplementary tables and figures

The tables in this section report quantities at a finer grain than the main paper, or quantities the main paper states only in prose or in a figure. The headline rejection-rate summaries are reported in Section S4, next to each data-generating process.

## Implied values, Simulation 1

The main paper describes the three forced-choice scenarios by the accuracy range they occupy. Table 13 gives the exact expected values behind those descriptions, and Table 14 gives the contrasts derived from them. The last row of the contrast table is the quantity the simulation is about: the age-by-group product term is exactly zero on the generating chance-corrected logit scale, while the group difference in expected accuracy still changes with age, by an amount that grows as the scenario moves toward the chance floor.

**Table 13**

*Expected accuracy implied by the three forced-choice scenarios, at the youngest, middle, and oldest age. Correct responses are out of k = 20 trials.*

| Scenario | Age | Group | Linear predictor | Expected accuracy | Correct out of 20 |
|---|---|---|---|---|---|
| Lower performance | 6 | Group 0 | -2.00 | 0.560 | 11.19 |
| Lower performance | 8 | Group 0 | -0.80 | 0.655 | 13.10 |
| Lower performance | 10 | Group 0 | 0.40 | 0.799 | 15.99 |
| Lower performance | 6 | Group 1 | -2.90 | 0.526 | 10.52 |
| Lower performance | 8 | Group 1 | -1.70 | 0.577 | 11.54 |
| Lower performance | 10 | Group 1 | -0.50 | 0.689 | 13.78 |
| Middle performance | 6 | Group 0 | -1.20 | 0.616 | 12.31 |
| Middle performance | 8 | Group 0 | 0.00 | 0.750 | 15.00 |
| Middle performance | 10 | Group 0 | 1.20 | 0.884 | 17.69 |
| Middle performance | 6 | Group 1 | -2.10 | 0.555 | 11.09 |
| Middle performance | 8 | Group 1 | -0.90 | 0.645 | 12.89 |
| Middle performance | 10 | Group 1 | 0.30 | 0.787 | 15.74 |
| Higher performance | 6 | Group 0 | -0.40 | 0.701 | 14.01 |

| Scenario | Age | Group | Linear predictor | Expected accuracy | Correct out of 20 |
|---|---|---|---|---|---|
| Higher performance | 8 | Group 0 | 0.80 | 0.845 | 16.90 |
| Higher performance | 10 | Group 0 | 2.00 | 0.940 | 18.81 |
| Higher performance | 6 | Group 1 | -1.30 | 0.607 | 12.14 |
| Higher performance | 8 | Group 1 | -0.10 | 0.738 | 14.75 |
| Higher performance | 10 | Group 1 | 1.10 | 0.875 | 17.50 |

**Table 14**

*Contrasts derived from the forced-choice scenarios. Probability points and correct-response units are on the observed accuracy scale; the link-scale column is on the generating chance-corrected logit scale.*

| Scenario | Contrast | Probability points | Correct out of 20 | Link scale |
|---|---|---|---|---|
| Lower performance | Group difference at youngest age: Group 1 minus Group 0 | -0.034 | -0.67 | – |
| Lower performance | Group difference at oldest age: Group 1 minus Group 0 | -0.111 | -2.21 | – |
| Lower performance | Age-related change in Group 0: oldest minus youngest | 0.240 | 4.79 | – |
| Lower performance | Age-related change in Group 1: oldest minus youngest | 0.163 | 3.25 | – |
| Lower performance | Change in group difference from youngest to oldest age | -0.077 | -1.54 | – |
| Lower performance | Generating link-scale age-by-group product term | – | – | 0.000 |

| Scenario | Contrast | Probability points | Correct out of 20 | Link scale |
|---|---|---|---|---|
| Middle performance | Group difference at youngest age: Group 1 minus Group 0 | -0.061 | -1.22 | – |
| Middle performance | Group difference at oldest age: Group 1 minus Group 0 | -0.097 | -1.94 | – |
| Middle performance | Age-related change in Group 0: oldest minus youngest | 0.269 | 5.37 | – |
| Middle performance | Age-related change in Group 1: oldest minus youngest | 0.233 | 4.65 | – |
| Middle performance | Change in group difference from youngest to oldest age | -0.036 | -0.72 | – |
| Middle performance | Generating link-scale age-by-group product term | – | – | 0.000 |
| Higher performance | Group difference at youngest age: Group 1 minus Group 0 | -0.094 | -1.87 | – |
| Higher performance | Group difference at oldest age: Group 1 minus Group 0 | -0.065 | -1.31 | – |
| Higher performance | Age-related change in Group 0: oldest minus youngest | 0.240 | 4.79 | – |
| Higher performance | Age-related change in Group 1: oldest minus youngest | 0.268 | 5.36 | – |
| Higher performance | Change in group difference from youngest to oldest age | 0.028 | 0.57 | – |
| Higher performance | Generating link-scale age-by-group product term | – | – | 0.000 |

### Deterministic link-scale decomposition, Simulation 2

The nominal rejection and pseudo-interaction detection rates for Simulation 2 are reported in Section S4. Table 15 instead reports the deterministic quantities behind them: for each generating-by-fitted link combination, the product-term contrast induced by evaluating the generating cell probabilities on the fitted link scale, and the corresponding response-scale difference-in-differences. Under a matched link the induced product term is zero by construction; under a wrong link it is a fixed nonzero value. The wrong-link rejection rate reflects power to detect that fitted-scale product term.

**Table 15**

*Deterministic link-scale decomposition for Simulation 2. The induced product term is evaluated on the fitted link scale; the response-scale difference-in-differences is on the probability scale.*

| Generating link | Fitted link | Match | Induced product (fitted scale) | Response-scale diff-in-diff |
|---|---|---|---|---|
| logit | logit | Matched link | 0.000 | -0.164 |
| logit | probit | Wrong link | -0.112 | -0.164 |
| probit | logit | Wrong link | 0.216 | -0.141 |
| probit | probit | Matched link | 0.000 | -0.141 |

### Implied values, Simulation 3

Table 16 gives the expected sum scores implied by the three sum-score scenarios, and Table 17 the contrasts derived from them. The generating latent-scale product term is zero in all three, yet the change in the group difference from low to high $x$, expressed in raw score points, is close to zero only in the middle of the scale and grows as the expected totals approach a bound. This is the compression the main paper describes, stated as a number rather than as a curve.

**Table 16**

*Expected sum scores implied by the three sum-score scenarios, at low, middle, and high values of the predictor. Totals run from 0 to 27.*

| Scenario | x | Group | Latent mean | Expected sum score | Percent of max |
|---|---|---|---|---|---|
| Lower range | -1 | Group 0 | -0.85 | 3.52 | 13.1% |
| Lower range | 0 | Group 0 | 0.00 | 6.20 | 23.0% |
| Lower range | 1 | Group 0 | 0.85 | 9.82 | 36.4% |
| Lower range | -1 | Group 1 | -1.75 | 1.75 | 6.5% |
| Lower range | 0 | Group 1 | -0.90 | 3.40 | 12.6% |
| Lower range | 1 | Group 1 | -0.05 | 6.01 | 22.3% |
| Middle range | -1 | Group 0 | -0.85 | 9.35 | 34.6% |
| Middle range | 0 | Group 0 | 0.00 | 13.50 | 50.0% |
| Middle range | 1 | Group 0 | 0.85 | 17.65 | 65.4% |
| Middle range | -1 | Group 1 | -1.75 | 5.65 | 20.9% |
| Middle range | 0 | Group 1 | -0.90 | 9.12 | 33.8% |
| Middle range | 1 | Group 1 | -0.05 | 13.25 | 49.1% |
| Upper range | -1 | Group 0 | -0.85 | 17.18 | 63.6% |
| Upper range | 0 | Group 0 | 0.00 | 20.80 | 77.0% |
| Upper range | 1 | Group 0 | 0.85 | 23.48 | 86.9% |
| Upper range | -1 | Group 1 | -1.75 | 12.75 | 47.2% |
| Upper range | 0 | Group 1 | -0.90 | 16.94 | 62.8% |
| Upper range | 1 | Group 1 | -0.05 | 20.61 | 76.3% |

**Table 17**

*Contrasts derived from the sum-score scenarios. Sum-score units and percent of maximum are on the observed total; the latent-scale column is on the generating scale.*

| Scenario | Contrast | Sum-score units | Percent of max | Latent scale |
|---|---|---|---|---|
| Lower range | Group difference at low x: Group 1 minus Group 0 | -1.774 | -6.6% | – |
| Lower range | Group difference at high x: Group 1 minus Group 0 | -3.807 | -14.1% | – |
| Lower range | x-related change in Group 0: high x minus low x | 6.295 | 23.3% | – |
| Lower range | x-related change in Group 1: high x minus low x | 4.262 | 15.8% | – |
| Lower range | Change in group difference from low x to high x | -2.034 | -7.5% | – |
| Lower range | Generating latent-scale x-by-group product term | – | – | 0.000 |
| Middle range | Group difference at low x: Group 1 minus Group 0 | -3.702 | -13.7% | – |
| Middle range | Group difference at high x: Group 1 minus Group 0 | -4.398 | -16.3% | – |
| Middle range | x-related change in Group 0: high x minus low x | 8.294 | 30.7% | – |
| Middle range | x-related change in Group 1: high x minus low x | 7.598 | 28.1% | – |
| Middle range | Change in group difference from low x to high x | -0.696 | -2.6% | – |
| Middle range | Generating latent-scale x-by-group product term | – | – | 0.000 |
| Upper range | Group difference at low x: Group 1 minus Group 0 | -4.432 | -16.4% | – |
| Upper range | Group difference at high x: Group 1 minus Group 0 | -2.863 | -10.6% | – |
| Upper range | x-related change in Group 0: high x minus low x | 6.295 | 23.3% | – |
| Upper range | x-related change in Group 1: high x minus low x | 7.865 | 29.1% | – |
| Upper range | Change in group difference from low x to high x | 1.570 | 5.8% | – |
| Upper range | Generating latent-scale x-by-group product term | – | – | 0.000 |

**Logit and probit fitted curves**

Section S4 notes that the within-family comparison is also illustrated with a separate grouped-binomial dataset of 1,000 observational units and 60 trials per unit. Figure 1 shows the two fitted curves, and Table 18 the corresponding coefficients and AIC values.

The illustration makes concrete why the two links are usually treated as interchangeable and why that is not sufficient for an interaction claim. Panel A shows fitted probabilities that are hard to tell apart by eye. Panel B plots their difference, which never exceeds about two probability points but is systematic rather than noisy: it changes sign across the predictor range, so the logit fit sits above the probit fit in some regions and below it in others. A difference between differences computed on one scale therefore cannot equal the same contrast computed on the other, which is the deterministic quantity tabulated in Table 15. The coefficients in Table 18 are on each model's own link scale and are not comparable across rows; the AIC values are comparable, and they differ by far more than the fitted probabilities do.

**Figure 1**

*Logit and probit models fitted to the same grouped-binomial dataset.*

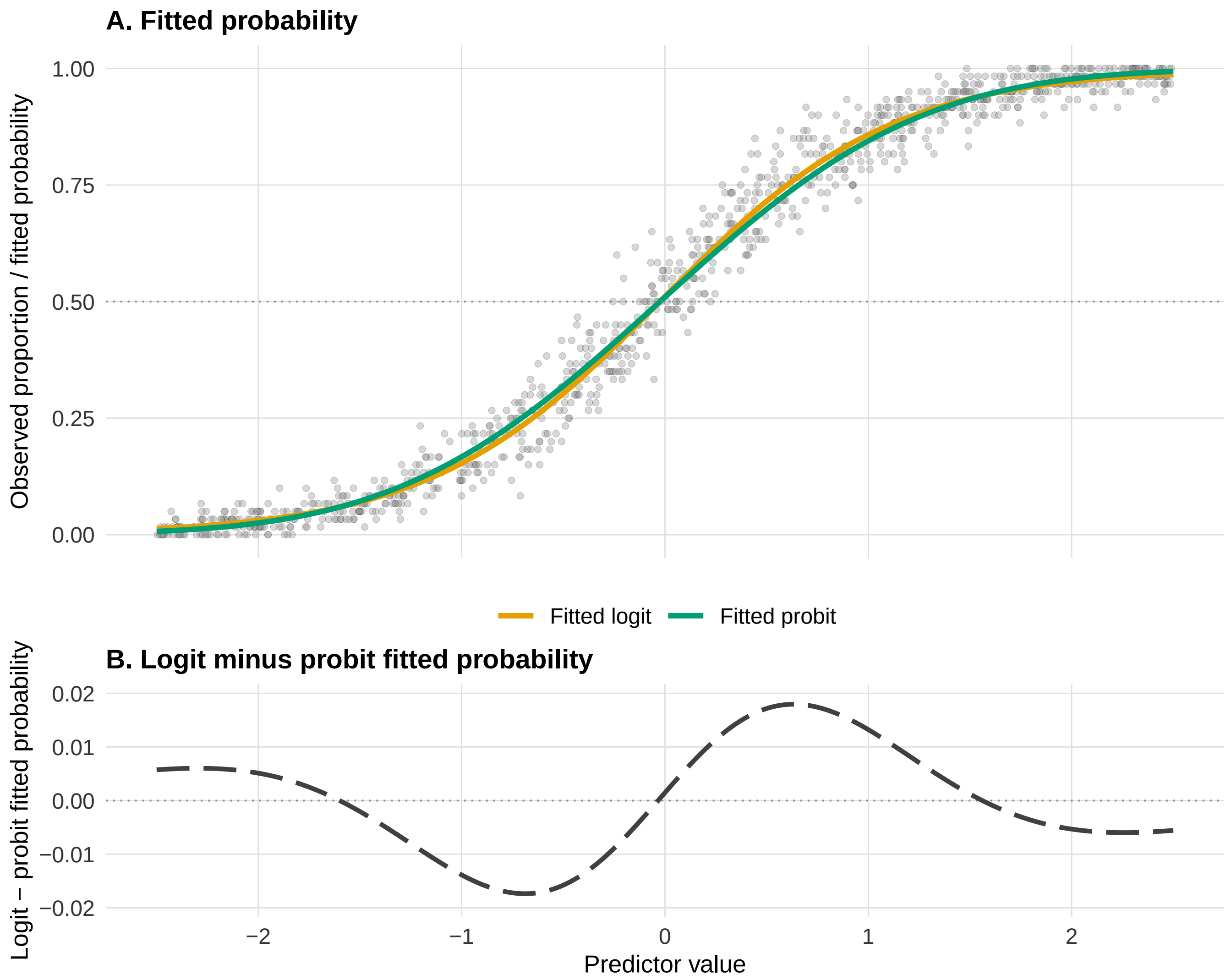


**Table 18**

*Coefficients and AIC for the logit and probit models fitted to the grouped-binomial illustration dataset. Intercept and slope are on each model's own link scale, so they are not directly comparable across rows.*

| Model | Intercept | Slope | AIC |
|---|---|---|---|
| logit | 0.045 | 1.753 | 4315.6 |
| probit | 0.024 | 0.990 | 4380.4 |

## Diagnostic detection rates and AIC strength

The main paper reports the DHARMa detection rate as a min-to-max range across checks. Table 19 gives the individual checks, the within-replication alignment between pseudo-interactions and applicable DHARMa checks, and the strength of the same-formula AIC comparison.

**Table 19**

*Diagnostic detection rates and AIC strength across five scenarios. D1 count errors (Poisson log to identity); D2 chance-floor accuracy (chance-corrected to standard logit); D3 binary 2x2 GLMM (probit to logit); D4 binary continuous-predictor GLMM (probit to logit); D5 Gamma mean response time (log to inverse).*

| Check | D1 | D2 | D3 | D4 | D5 |
|---|---|---|---|---|---|
| Pseudo-interaction (wrong link) | 2682/2798 (95.9%) | 1733 (57.8%) | 301 (10.0%) | 201 (6.7%) | 905 (30.2%) |
| DHARMa: uniformity | 0.9% | 0.0% | 3.1% | 2.9% | 0.0% |
| DHARMa: dispersion | 22.7% | 1.4% | 0.0% | 0.0% | 0.1% |
| DHARMa: quantiles vs fitted | 63.3% | 1.9% | 2.5% | 2.3% | 2.7% |
| DHARMa: quantiles vs predictor | 90.6% | 6.0% | – | 2.1% | 7.3% |

| Check | D1 | D2 | D3 | D4 | D5 |
|---|---|---|---|---|---|
| DHARMa: across design cells | – | – | 0.4% | – | – |
| Pseudo-interactions missed by all applicable DHARMa checks | 192/2682 (7.2%) | 1589/1733 (91.7%) | 283/301 (94.0%) | 191/201 (95.0%) | 827/905 (91.4%) |
| Pregibon link check | 99.3% | 28.8% | N.A. | 4.7% | 32.7% |
| AIC favors target link | 99.1% | 77.6% | 95.4% | 95.5% | 78.3% |
| AIC favors misspecified link | 0.9% | 22.4% | 4.6% | 4.5% | 21.7% |
| Median <U+0394>AIC | 22.7 (n = 2798) | 2.2 (n = 3000) | 5.8 (n = 3000) | 4.9 (n = 3000) | 2.4 (n = 3000) |
| \|<U+0394>AIC\| < 2 | 39/2798 (1.4%) | 1189 (39.6%) | 351 (11.7%) | 462 (15.4%) | 1108 (36.9%) |

All rates are based on 3,000 usable replications unless otherwise noted. “Missed” means that none of the applicable prespecified DHARMa checks had $p < .05$; pseudo-interactions for which all five DHARMa p-values were missing are excluded from this conditional denominator. $\Delta$AIC = AIC(wrong link) – AIC(target link), so positive values favor the target link and negative values favor the wrong link. The sample size shown with the median and the denominator for $|\Delta\text{AIC}| < 2$ are the finite AIC comparisons. N.A. indicates that the Pregibon-style added term is not identifiable in the saturated $2 \times 2$ interaction model.

**Distribution of the AIC difference**

The main paper argues that the proportion of replications in which AIC favours the target link is not informative on its own, because winning by less than one AIC unit and winning by fifty describe different situations. Table 20 therefore gives the distribution of $\Delta$AIC rather than only its median. The contrast between the count scenario and the chance-floor scenario is the clearest: both favour the target link in most replications, but the count scenario separates the two links by tens of AIC units across the whole distribution, whereas in the chance-floor scenario the lower decile is negative, meaning that in a non-trivial share of replications AIC actively preferred the misspecified link, and roughly two fifths of the comparisons fall within two units of a tie.

**Table 20**

*Distribution of the AIC difference across replications, by diagnostic scenario. Delta AIC = AIC(wrong link) - AIC(target link), so positive values favor the target link. n = replications with both AIC values available.*

| Scenario | n | 10th | 25th | Median | 75th | 90th | Within 2 units | Favors target |
|---|---|---|---|---|---|---|---|---|
| Count errors | 2798 | 10.4 | 16.2 | 22.7 | 29.4 | 35.9 | 1.4% | 99.1% |
| Chance-floor accuracy | 3000 | -1.6 | 0.2 | 2.2 | 4.3 | 6.4 | 39.6% | 77.6% |
| Binary repeated trials | 3000 | 1.4 | 3.5 | 5.8 | 8.2 | 10.4 | 11.7% | 95.4% |
| Binary continuous predictor | 3000 | 1.1 | 2.9 | 4.9 | 7.0 | 8.6 | 15.4% | 95.5% |
| Gamma mean response time | 3000 | -1.5 | 0.3 | 2.4 | 4.5 | 6.5 | 36.9% | 78.3% |

## Calibration under correctly specified links

The same DHARMa and Pregibon-style checks were applied to the corresponding correct-link interaction model in each replication. These baseline flagging rates provide a reference for interpreting detection under the wrong link. They reuse the same simulated datasets and scenarios rather than defining a new simulation.

The correct-link column provides the baseline needed to interpret detection under the wrong link. The four DHARMa checks are well calibrated throughout. For the Pregibon-style check, the random-intercept GLMMs use the population-level rather than subject-specific fitted linear predictor. The $2 \times 2$ version is not applicable because its saturated fixed-effects design already spans the proposed squared term; the continuous-predictor version is estimable, and its wrong-link flagging rate should be read against its correct-link calibration rate.

**Table 21**

*Diagnostic calibration under correctly specified and misspecified links. Each cell gives the flagging rate with its Wilson 95% confidence interval and the number of successful diagnostic fits. Each cell had 3,000 attempted replications.*

| Scenario | Diagnostic | Correct-link flagging | Wrong-link flagging |
|---|---|---|---|
| Count errors | DHARMa uniformity | 0.0% [0.0, 0.2] | 0.9% [0.6, 1.4] (n = 2798) |
| Count errors | DHARMa dispersion | 5.2% [4.5, 6.1] | 22.7% [21.1, 24.2] (n = 2798) |
| Count errors | DHARMa residual quantiles over fitted values | 3.1% [2.5, 3.8] | 63.3% [61.5, 65.0] (n = 2798) |
| Count errors | DHARMa residual quantiles over focal predictor | 4.0% [3.4, 4.8] | 90.6% [89.4, 91.6] (n = 2798) |
| Count errors | Pregibon-style added-term link check | 5.4% [4.6, 6.2] | 99.3% [98.9, 99.5] (n = 2796) |
| Chance-floor accuracy | DHARMa uniformity | 0.0% [0.0, 0.2] | 0.0% [0.0, 0.2] |
| Chance-floor accuracy | DHARMa dispersion | 1.3% [1.0, 1.8] | 1.4% [1.0, 1.9] |
| Chance-floor accuracy | DHARMa residual quantiles over fitted values | 1.1% [0.8, 1.5] | 1.9% [1.5, 2.5] |
| Chance-floor accuracy | DHARMa residual quantiles over focal predictor | 1.9% [1.5, 2.5] | 6.0% [5.2, 6.9] |
| Chance-floor accuracy | Pregibon-style added-term link check | 4.1% [3.4, 4.8] | 28.8% [27.2, 30.4] |
| Binary repeated trials | DHARMa uniformity | 4.2% [3.5, 4.9] | 3.1% [2.5, 3.7] |
| Binary repeated trials | DHARMa dispersion | 0.0% [0.0, 0.1] | 0.0% [0.0, 0.1] |
| Binary repeated trials | DHARMa residual quantiles over fitted values | 3.7% [3.1, 4.4] | 2.5% [2.0, 3.1] |
| Binary repeated trials | DHARMa residual distribution across design cells | 0.9% [0.6, 1.3] | 0.4% [0.3, 0.7] |
| Binary repeated trials | Pregibon-style added-term link check | N.A. | N.A. |
| Binary continuous predictor | DHARMa uniformity | 3.9% [3.3, 4.7] | 2.9% [2.4, 3.6] |
| Binary continuous predictor | DHARMa dispersion | 0.0% [0.0, 0.1] | 0.0% [0.0, 0.1] |

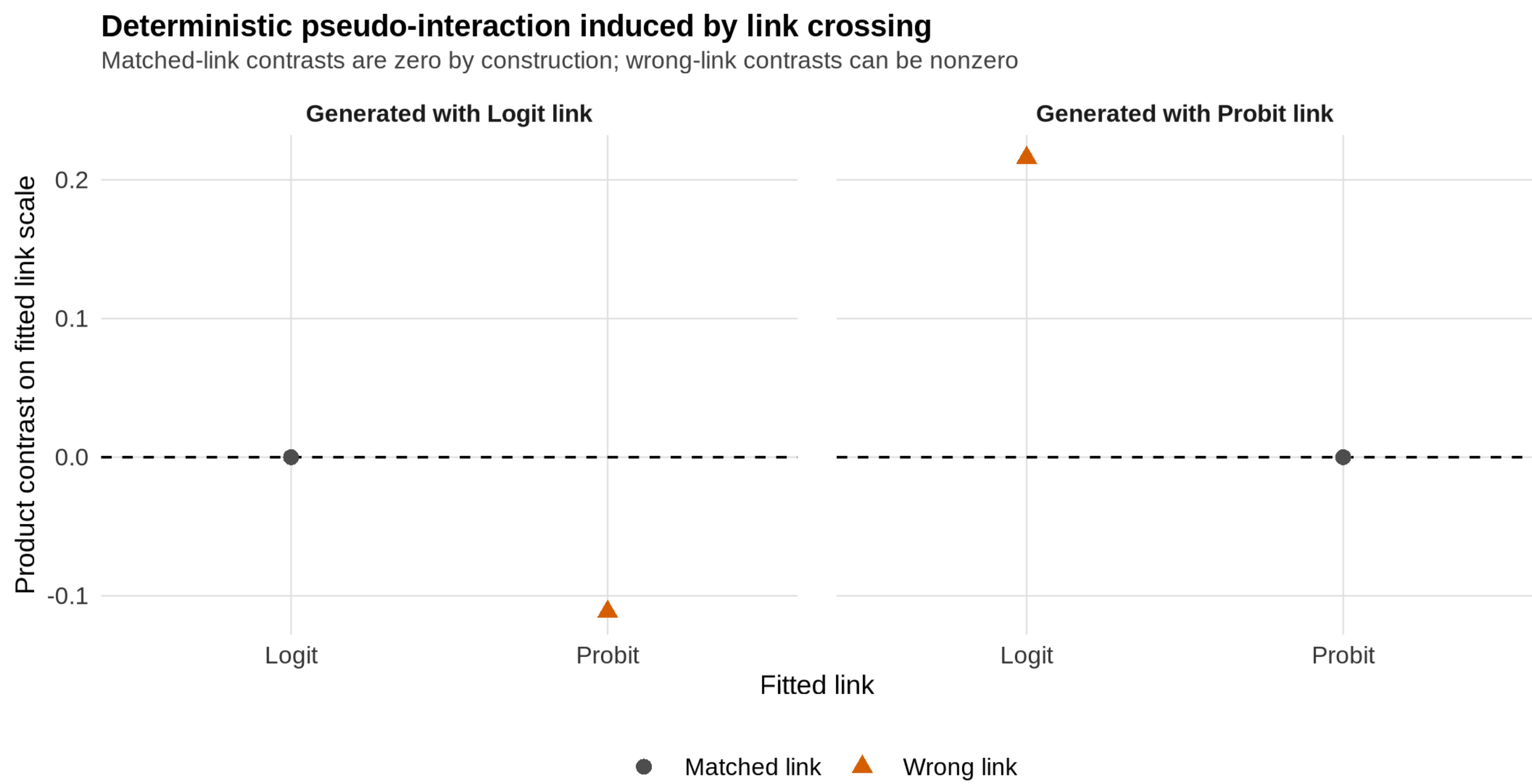
| Scenario | Diagnostic | Correct-link flagging | Wrong-link flagging |
|---|---|---|---|
| Binary continuous predictor | DHARMa residual quantiles over fitted values | 2.8% [2.2, 3.4] | 2.3% [1.9, 2.9] |
| Binary continuous predictor | DHARMa residual quantiles over focal predictor | 2.6% [2.1, 3.2] | 2.1% [1.7, 2.7] |
| Binary continuous predictor | Pregibon-style added-term link check | 4.4% [3.7, 5.2] | 4.7% [4.0, 5.5] |
| Gamma mean response time | DHARMa uniformity | 0.0% [0.0, 0.1] | 0.0% [0.0, 0.1] |
| Gamma mean response time | DHARMa dispersion | 0.2% [0.1, 0.4] | 0.1% [0.0, 0.2] |
| Gamma mean response time | DHARMa residual quantiles over fitted values | 1.7% [1.3, 2.2] | 2.7% [2.2, 3.3] |
| Gamma mean response time | DHARMa residual quantiles over focal predictor | 2.1% [1.6, 2.7] | 7.3% [6.5, 8.3] |
| Gamma mean response time | Pregibon-style added-term link check | 5.4% [4.7, 6.3] | 32.7% [31.1, 34.4] |

**Deterministic pseudo-interaction under the wrong link**

Figure 2 shows the deterministic pseudo-interaction induced on the wrong link scale in Simulation 2, before any sampling noise. It makes explicit why matched-link cells imply an exactly additive structure while wrong-link cells do not.

**Figure 2**

*Deterministic pseudo-interaction induced on the wrong link scale in Simulation 2.*




## S6. Data, code, and computational environment

All analyses were conducted in R, and the manuscript and this supplement were written in Quarto. Each simulation sets a fixed random seed, and the number of Monte Carlo replications and the significance level are recorded together with the saved results. Exact package versions are pinned with `renv`, which is the source of truth for the computational environment; the key modelling dependencies are `glmmTMB` for the mixed-model simulations and `DHARMa` for the diagnostic checks. The complete data, analysis code, preregistration materials, and generated outputs are archived in the public repository associated with the article, so that every table and figure can be regenerated from the raw inputs.

The manuscript simulations use a small number of transparent scenarios. A broader precomputed sensitivity atlas generated in `simulation-atlas/` varies key parameters around those anchors, and the Shiny app explores its compact summaries without running simulations interactively. Replication-level atlas results are intended for the public repository or OSF archive and are not required to render this supplement.

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